BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SWAT

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2022

MATHEMATICS

Past Paper 2022 • Paper Code: 102201

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 15

Note: 1) Attempting all MCQs is compulsory. This paper along with the OMR sheet must be returned to the superintendent after due time. 2) Fill the circle A B C D, which one is correct with blue or black ball point in separate OMR Sheet like ● 3) If more than one circle in the OMR sheet is filled then no credit will be given to such answer.

Q1. In the given set of data 5,7,7,5,3,7,2,8,2 mode is.

(A) 9
(B) 5
(C) 2
(D) 7

Q2. The formula Σx/n determines.

(A) Arithmetic Mean
(B) Median
(C) Mode
(D) G.M

Q3. If Sinθ > 0, tan θ > 0, then θ lies in _____ quadrant.

(A) IV
(B) III
(C) II
(D) I

Q4. If an object is above the level of observer then the angle formed between the horizontal line and observer's line of sight is called _____.

(A) Angle of depression
(B) Angle of elevation
(C) Obtuse angle
(D) None of these

Q5. Apollonius was born in _____.

(A) Perga
(B) Istanbul
(C) Islamabad
(D) Kabul

Q6. If the distance between the centres of two circles is equal to the sum of their radii, then circles will.

(A) Intersect
(B) Do not intersect
(C) Touch each other externally
(D) Touch each other internally

Q7. The measure of the external angle of a regular hexagon is.

(A) π/3
(B) π/4
(C) π/6
(D) None of these

Q8. If two chords of a circle are equal, then their corresponding arcs are _____.

(A) Congruent
(B) Not congruent
(C) Concurrent
(D) None of these

Q9. If (x +1) (x -5) = 0, then the solutions are.

(A) x =1,-5
(B) x =1, 5
(C) x =-1, -5
(D) x =-1, 5

Q10. For the quadratic equation $ax^2+bx+c=0$ if $b^2-4ac=0$, then roots are _____.

(A) Unequal
(B) Imaginary
(C) Equal
(D) None of these

Q11. If α , β are the roots of $9x^2-27x+k=0$ then α β = _____.

(A) 27/9
(B) K/9
(C) -K/9
(D) -27/9

Q12. The third proportional of x and y is.

(A) xy
(B) x/y
(C) y^2/x
(D) None of these

Q13. If a : b = c : d then (a+b) : b :: (c+d) : d it is _____ property.

(A) Componendo
(B) Dividendo
(C) Invertendo
(D) Alternendo

Q14. $(x^2+2)/(x^2+2x+2)$ is _____.

(A) Proper fraction
(B) Improper fraction
(C) Irrational fraction
(D) None of these

Q15. If A={3,4,5}, B={5,6,7} then A ∩ B = _____.

(A) {7}
(B) {4,6}
(C) {5}
(D) {3,7}

SECTION "B"

Time Allowed: 2 hours and 40 minutes | **Maximum Marks:** 36

QII. Attempt any NINE Parts out of the following. Each Part carries equal marks.

(i) Solve $2x + 4/x = 9$.

(ii) Prove that $(-1+\iota\sqrt{3})^4 (-1-\iota\sqrt{3})^3 = 512\ w^2$.

(iii) The sum of two numbers is 11 and sum of their square is 65. Find the numbers.

(iv) If 3x-1, 4, 35 are in continued proportion, find the value of x.

(v) If a : b = c : d then prove that (2a+3b)/(2a-3b) = (2c+3d)/(2c-3d).

(vi) Resolve 1/((x+1)(x+2)) into partial fractions.

(vii) Verify associative property of intersection for the following sets. A={1,2,3,....,100} B={2,4,6,...,100} C={1,3,5,...,99}

(viii) If x= {1,2,3,4}, y={5,6,7,8} then write a function which is onto from y to x.

(ix) What is the range of the data 295, 258, 270, 311, 311?

(x) Find the area of sector of a circle whose radius is 4m, with central angle 12 radian.

(xi) Prove that Sin x/(1+Cos x) + (1+Cos x)/Sin x = 2 Cosec x.

(xii) A building that is 21 meters tall casts a shadow 25 meters long. Find the angle of elevation of the sun to the nearest degree.

SECTION "C"

Maximum Marks: 24

Note: Attempt any THREE questions of the following. Each question carries equal Marks.

QIII. Prove that: One and only one circle can pass through three non-collinear points.

QIV. The two tangents drawn to a circle from a point outside it, are equal in length.

QV. If two chords of a circle are equal, then their corresponding arcs are congruent.

QVI. Draw a square of 6cm. Circumscribe a circle about that square and then inscribe a circle in the same square.